

Research Papers and Journals

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 |
Research/Research_Papers_and_Journals.md

All academic sources used in the Resilio FYP report, organised by research theme. Full citations in Harvard style are in Chapter 6 of the report. This file provides a thematic map of the literature to show which sources informed which design and development decisions.

THEME 1: Mental Health in Nepal — Epidemiology and System

These sources established the evidence base for the problem statement in Chapter 1 and Appendix A.

Source	Key Finding Used	Used In
Ministry of Health and Population, Nepal (2020) *Nepal Mental Health Survey 2020*	12.9% lifetime prevalence; 92% treatment gap; workforce shortage statistics	Ch. 1, Ch. 5, Appendix A
World Health Organization (2021) *WHO Special Initiative for Mental Health: Situational Assessment — Nepal*	Confirms treatment gap; workforce density figures	Ch. 1, Appendix A
Luitel, N.P. et al. (2015) 'Mental health care in Nepal: Current situation and challenges' *Conflict and Health*	Rural service distribution; barriers; district-level care gaps	Ch. 1, Appendix A, B
Atreya, A., Upreti, M. and Nepal, S. (2023) 'Barriers to mental health care access in Nepal' *International Journal of Social Psychiatry*	Updated barriers data: cost, distance, stigma percentages	Ch. 1, Appendix A
Jordans, M.J.D. et al. (2016) 'Development and pilot testing of a mental healthcare plan in rural Nepal' *British Journal of Psychiatry*	Rural access inequity: 65% services concentrated in Kathmandu	Ch. 1, Appendix A, B

Source	Key Finding Used	Used In
Kohrt, B.A. and Harper, I. (2008) 'Navigating diagnoses: Understanding mind-body relations, mental health, and stigma in Nepal' *Culture, Medicine and Psychiatry*	Cultural stigma dynamics; traditional healer preference; help-seeking barriers	Ch. 1, Ch. 5, Appendix A, B
The Lancet Regional Health Southeast Asia (2025) 'Nepal's mental health system from public health perspective'	Most recent systemic assessment; confirms ongoing workforce crisis	Ch. 1, Appendix A
Kathmandu Post (2023) 'Nepal's Mental Health Crisis'	Media evidence of public awareness of the crisis	Ch. 1
Kathmandu Post (2024) 'More Farmers and Students are Killing Themselves in Nepal'	Suicide rate trends; risk demographic data	Ch. 1
Bhandari, P. et al. (2024) 'Exploring trends: Five-year analysis of suicide rates in Nepal' *Mental Illness*	Quantitative suicide trend analysis over 5 years	Ch. 1
Aawaaj News (2025) 'Adolescent Mental Health in Nepal: Rising Risks, Stigma, and the Need for Early Support'	Adolescent-specific mental health risk data	Ch. 1
Poudel, B. et al. (2023) 'Recurrent COVID-19 waves and lockdown: Impact on daily life and mental health' *Mental Illness*	Nepal-specific COVID-19 mental health impact	Ch. 1

THEME 2: Digital Mental Health Platforms — Evidence and Effectiveness

These sources informed the literature review in Chapter 2 and the design rationale in Chapter 3.

Source	Key Finding Used	Used In
Bakker, D. et al. (2016) 'Mental health smartphone apps: Review and evidence-based recommendations' *JMIR Mental Health*	Fragmentation as primary engagement failure; integration improves outcomes	Ch. 2, Ch. 5, Appendix B

Source	Key Finding Used	Used In
Fleming, T. et al. (2019) 'Beyond the trial: Systematic review of real-world uptake and engagement with digital self-help interventions' *Journal of Medical Internet Research*	60% higher retention with integrated platforms; gamification impact	Ch. 2, Ch. 5, Appendix B, D
Chandrashekar, P. (2018) 'Do mental health mobile apps work' *mHealth*	Evidence-based recommendations for app design; streak avoidance; professional integration	Ch. 2, Ch. 5, Appendix B, D
Naslund, J.A. et al. (2017) 'Digital technology for treating and preventing mental disorders in LMICs' *The Lancet Psychiatry*	Digital platforms as solution for low-resource mental health systems	Ch. 2
Torous, J. et al. (2019) 'Digital mental health and COVID-19' *JMIR Mental Health*	Accelerated adoption post-COVID; wearable integration potential	Ch. 2, Ch. 5
Inkster, B., Sarda, S. and Subramanian, V. (2018) 'An empathy-driven, conversational AI agent (Wysa) for digital mental well-being' *JMIR mHealth and uHealth*	Wysa evidence base; AI chatbot effectiveness	Ch. 2
Abd-Alrazaq, A.A. et al. (2020) 'An overview of the features of chatbots in mental health: A scoping review' *International Journal of Medical Informatics*	Chatbot features; AI matching potential	Ch. 2, Ch. 5, Appendix N
Naslund, J.A. et al. (2016) 'The future of mental health care: Peer-to-peer support and social media' *Epidemiology and Psychiatric Sciences*	Peer support community evidence	Ch. 5, Appendix N
Linardon, J. and Fuller-Tyszkiewicz, M. (2020) 'Attrition and adherence in smartphone-delivered interventions' *Journal of Consulting and Clinical Psychology*	26% first-week drop-off rate in mental health apps	Ch. 2, Appendix B

THEME 3: Clinical Measurement Instruments

These sources validate the PHQ-9 and GAD-7 instruments used in Resilio.

Source	Key Finding Used	Used In
Kroenke, K., Spitzer, R.L. and Williams, J.B.W. (2001) 'The PHQ-9: Validity of a brief depression severity measure' *Journal of General Internal Medicine*	PHQ-9 validation; scoring bands; clinical threshold justification	Ch. 3, Ch. 5, Appendix D, M
Spitzer, R.L. et al. (2006) 'A brief measure for assessing generalized anxiety disorder: The GAD-7' *Archives of Internal Medicine*	GAD-7 validation; scoring interpretation	Ch. 3, Ch. 5, Appendix D, M
Shiffman, S., Stone, A.A. and Hufford, M.R. (2008) 'Ecological momentary assessment' *Annual Review of Clinical Psychology*	EMA principles underpinning mood logging design	Ch. 3, Ch. 5, Appendix D

THEME 4: Mindfulness and Wellness App Evidence

These sources support the content hub and wellness game design rationale.

Source	Key Finding Used	Used In
Larochette, A., Churchill, E. and DuBois, J.M. (2024) 'Mind the app: More time spent on Headspace leads to beneficial day-to-day changes' *Digital Health*	Headspace effectiveness: mindfulness, depression, anxiety improvements	Ch. 2
Vago, D.R. et al. (2024) 'The real-world impact of app-based mindfulness on Headspace members with moderate and severe perceived stress' *JMIR mHealth and uHealth*	Real-world stress reduction with app-based mindfulness	Ch. 2

THEME 5: Telepsychology / Video Consultation

Source	Key Finding Used	Used In
Backhaus, A. et al. (2012) 'Videoconferencing psychotherapy: A systematic review' *Psychological Services*	Clinical equivalence of video therapy vs in-person; evidence base for WebRTC feature	Ch. 2, Ch. 3
Patel, V. et al. (2018) 'The Lancet Commission on global mental health and sustainable development' *The Lancet*	Global cost of untreated mental illness; investment return data	Ch. 1, Ch. 2
World Health Organization (2016) *Investing in Treatment for Depression and Anxiety Leads to Fourfold Return*	USD 4 return per USD 1 invested; economic justification	Ch. 1, Ch. 2

THEME 6: Software Engineering Methodology

Source	Key Finding Used	Used In
Kruchten, P. (2003) *The Rational Unified Process: An Introduction*	RUP framework: four phases, workflows, artefacts	Ch. 3, Appendix E
Martin, R.C. (2017) *Clean Architecture: A Craftsman's Guide to Software Structure and Design*	Clean Architecture layers: entities, use cases, interfaces, infrastructure	Ch. 3
Beck, K. et al. (2001) *Manifesto for Agile Software Development*	Agile principles (used in methodology comparison)	Ch. 3, Appendix E
Sommerville, I. (2016) *Software Engineering*. 10th edn	Waterfall model; software quality attributes; testing frameworks	Ch. 3, Ch. 4, Appendix E
Pressman, R.S. (2014) *Software Engineering: A Practitioner's Approach*. 8th edn	Software testing principles; system testing framework	Ch. 4, Appendix E
Martin, J. (1991) *Rapid Application Development*	RAD/Prototype methodology description	Appendix E

THEME 7: Nepal Digital Infrastructure and Market

Source	Key Finding Used	Used In
DataReportal (2025) *Digital 2025: Nepal*	Internet penetration; smartphone usage; demographic data	Ch. 1, Ch. 2
Nepal Telecommunications Authority (2022) *Management Information System Report*	30% without regular internet; coverage statistics	Ch. 1, Ch. 5, Appendix B, D
World Bank (2023) *Nepal GDP per Capita*	GDP per capita USD 1,336; pricing context	Ch. 1
Grand View Research (2022) *Mental Health Apps Market Size*	Global market size and growth rate	Ch. 2

THEME 8: Platform Comparisons (BetterHelp, Headspace, Wysa)

Source	Key Finding Used	Used In
BusinessWire (2025) 'BetterHelp Surpasses 5 Million People Benefiting from Online Therapy Service'	BetterHelp user base size; market penetration	Ch. 2, Appendix C
BetterHelp (2024) *BetterHelp Platform Quality and Outcomes in 2024*	BetterHelp pricing and outcomes data	Ch. 2, Appendix C
Headspace (2024) *Headspace: Meditation and Sleep*	Headspace feature set and pricing	Ch. 2, Appendix C
Wysa (2024) *Wysa Clinical Evidence and Research*	82% crisis detection rate; AI chatbot evidence	Ch. 2, Ch. 5, Appendix C
HamroPatro (2024) *Hamro Patro Health*	Local competitor feature analysis	Ch. 2, Appendix C

THEME 9: Legal, Privacy, and Compliance

Source	Key Finding Used	Used In
Government of Nepal (2018) *Individual Privacy Act 2018*	Privacy obligations for health data collection in Nepal	Ch. 5
Ministry of Communication and Information Technology, Nepal (2023) *Draft National Cybersecurity Policy*	Data residency considerations	Ch. 5
Apple Inc. (2024) *App Store Review Guidelines: Health and Medical Apps*	iOS compliance requirements for health platforms	Ch. 5, Appendix N
SuperTokens (2024) *SuperTokens Documentation: Passwordless Recipe*	Authentication implementation reference	Ch. 3, Appendix I

THEME 10: Future Work Research

Source	Key Finding Used	Used In
Coppersmith, G., Dredze, M. and Harman, C. (2014) 'Quantifying mental health signals in Twitter' *ACL Workshop*	NLP for mood analysis — future feature basis	Ch. 5, Appendix N
Guntuku, S.C. et al. (2017) 'Detecting depression and mental illness on social media' *Current Opinion in Behavioral Sciences*	NLP depression detection — future feature evidence	Ch. 5, Appendix N
Firth, J. et al. (2017) 'The efficacy of smartphone-based mental health interventions for depressive symptoms' *World Psychiatry*	Wearable integration evidence	Ch. 5, Appendix N
Yalom, I.D. and Leszcz, M. (2020) *The Theory and Practice of Group Psychotherapy*. 6th edn	Group therapy evidence — future WebRTC extension	Ch. 5, Appendix N
Miner, A.S. et al. (2016) 'Smartphone-based conversational agents' *JAMA Internal Medicine*	AI chatbot capabilities	Ch. 5, Appendix N

Source	Key Finding Used	Used In
Fortuna, K.L. et al. (2019) 'Exploring the scope of peer support' *Psychiatric Rehabilitation Journal*	Peer support communities — future feature	Ch. 5, Appendix N
Morin, C.M. and Espie, C.A. (2003) *Insomnia: A Clinical Guide to Assessment and Treatment*	Sleep content rationale	Background reading

COVID-19 and Mental Health Context

Source	Key Finding Used	Used In
World Health Organization (2022) 'COVID-19 Pandemic Triggers 25% Increase in Prevalence of Anxiety and Depression Worldwide'	Global context for increased demand	Ch. 1
World Health Organization (2023) *Depressive Disorder (Depression)*	WHO definition and global prevalence update	Ch. 1, Ch. 2
Sigdel, A. et al. (2021) 'Depression, anxiety and depression-anxiety comorbidity amid COVID-19 pandemic: A systematic review' *Journal of Psychiatric Research*	Nepal COVID-19 mental health impact	Ch. 1